



# “A COMPARATIVE STUDY OF DEEP LEARNING MODELS FOR PREDICTING E-COMMERCE REVENUE”

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ROLL NO.:38





# I. PROJECT OVERVIEW

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- This project focuses on **predicting e-commerce revenue** using deep learning models:
- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)
- Long Short-Term Memory (LSTM)
- The goal is to compare these models and identify which performs best.





## 2. DATASET DESCRIPTION

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- The dataset contains **5000 records of e-commerce sales**
- Includes numerical features like sales, orders, etc.
- Target variable: **Revenue**

### ◆ 3. DATA PREPROCESSING

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Steps performed:

- Removed unnecessary columns
  - Separated features (X) and target (y)
  - Applied **train-test split (80/20)**
  - Used **StandardScaler** for normalization
- 👉 This improves model performance and stability.





## 4. MODELS USED



### ✓ ANN (Artificial Neural Network)

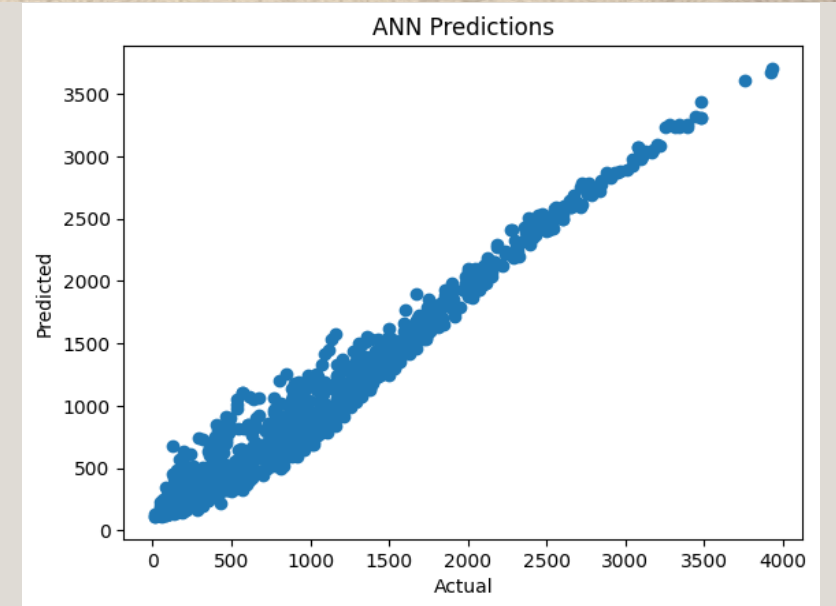
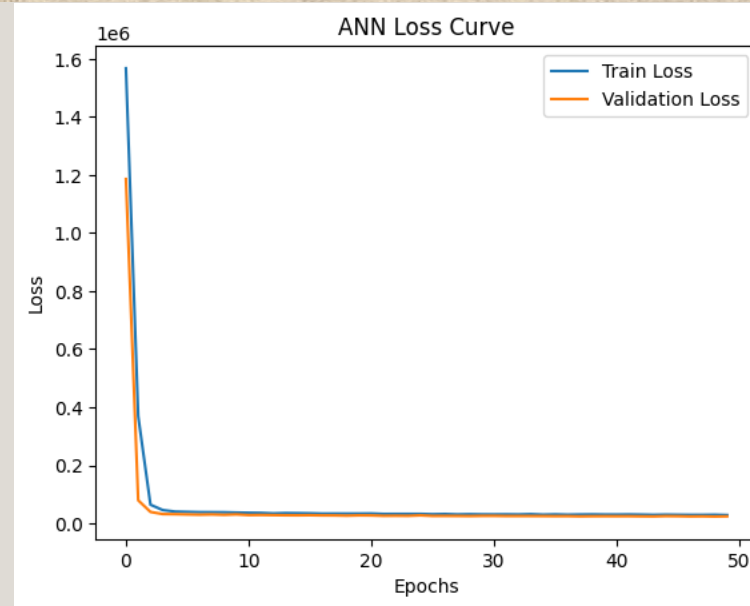
Best suited for **structured/tabular data**

Learns complex relationships between features

#### **Result:**

Lowest error

Best prediction accuracy





## 4. MODELS USED



### ✓ CNN

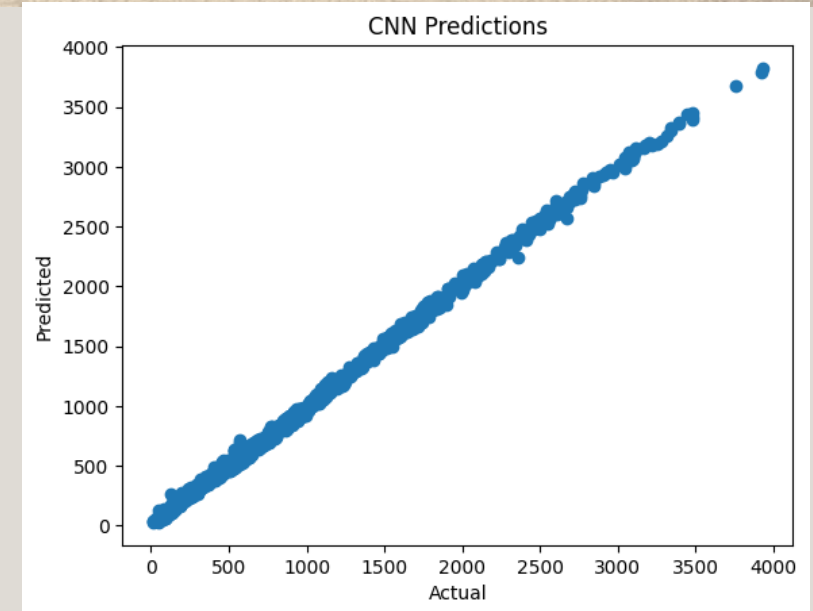
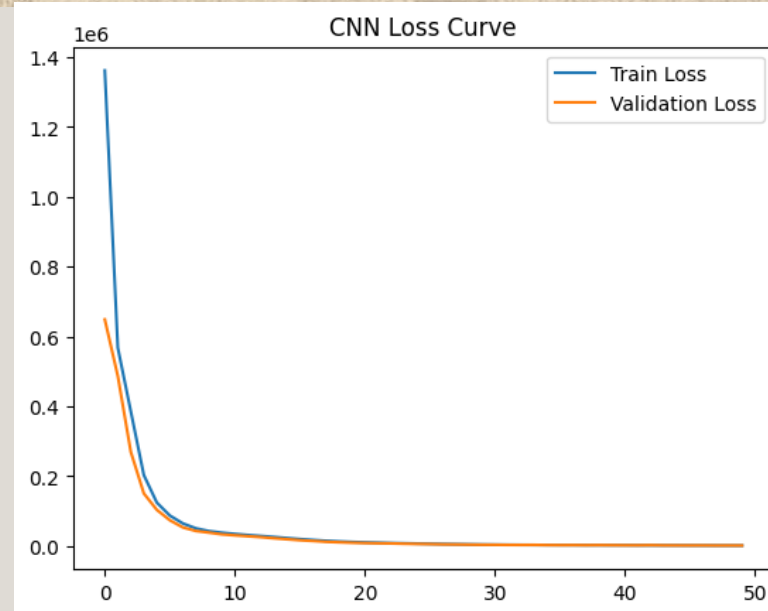
Extracts **pattern-based features**

More suitable for images or spatial data

#### **Result:**

Moderate performance

Not ideal for tabular dataset







## 4. MODELS USED



### ✓ LSTM

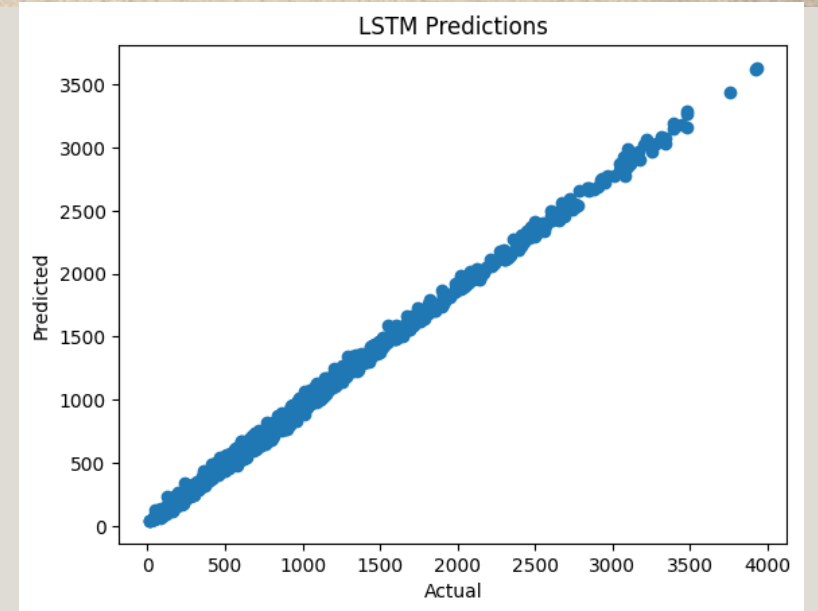
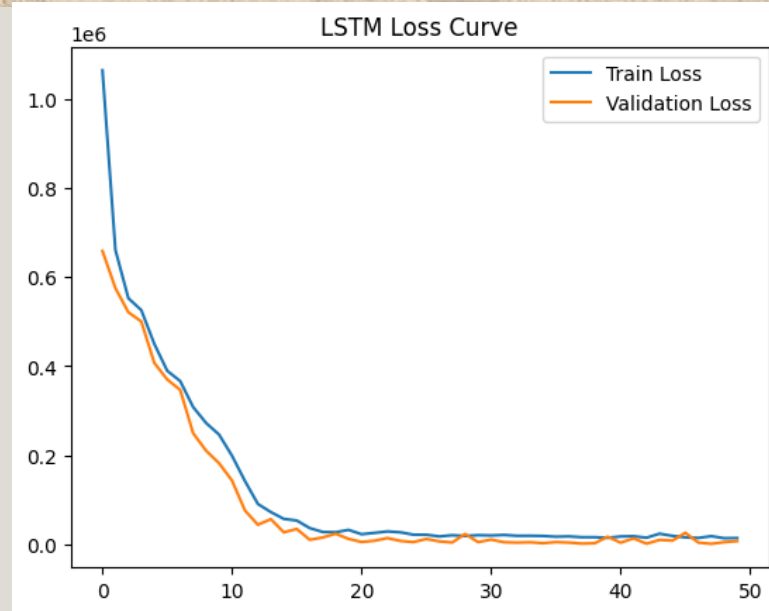
Designed for **time-series data**

Captures past dependencies

#### Result:

Performs well only if data has sequence

Not very effective here





## 5. GRAPH EXPLANATION

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-  **Loss Curve**
  - Shows error reduction during training
  - ANN converges faster → better learning
-  **Actual vs Predicted Graph**
  - Points close to diagonal line = good prediction
  - ANN graph shows **closer alignment**



## ◆ 6. FINAL INTERPRETATION

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**ANN performed best because:**

Dataset is structured

No strong sequential pattern

**CNN and LSTM are less suitable because:**

CNN is for spatial data

LSTM is for time-series

👉 **Conclusion: ANN is the best model for this dataset**



## ◆ 7. CONCLUSION

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“Among ANN, CNN, and LSTM models, ANN achieved the highest accuracy and lowest error, making it the most suitable model for revenue prediction on structured e-commerce data.”